Khue (Cherry) Pham

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Education

Northeastern University, The Roux Institute – M.S. Computer Science Dec 2027

Relevant Coursework: Programming Design Paradigms; Machine Learning; Database Management Systems

Olin College of Engineering – B.S. Electrical and Computer Engineering May 2025

Relevant Coursework: Advanced Algorithms; Software Systems; Software Design; Data Structures and Algorithms; Neurotechnology; Computer Architecture; Modeling and Simulation; Analog/Digital Communication

Technical Skills

Languages: SQL, Python

Software Engineering: REST APIs, FastAPI, PostgreSQL, Docker, Distributed Systems, System Design, SQLAlchemy, AWS, CI/CD, Git

ML/AI: pgvector, PyTorch, Hugging Face Transformers, LLMs, Fine-tuning, semantic search

Experience

Florida Center for Reading Research Remote

Machine Learning Engineer Sep 2024 – Aug 2025

- Designed a hybrid SQL/NoSQL data model from an ERD, using relational storage for interconnected student, teacher, parent, classroom, and game data and document storage for variable per-game metrics; deployed on Red Hat OpenShift with OAuth 2.0 and role-based access control.
- Built RAG pipelines with pgvector-backed semantic search (PostgreSQL), improving retrieval accuracy by 15–25% on children's reading development tasks; evaluated using BLEU/ROUGE benchmarks.
- Deployed LLM inference behind a REST API endpoint serving an education portal, with scalable cloud infrastructure supporting SQL/NoSQL storage and OAuth-based authentication for multi-user research access.

Air Partners Needham, MA

Data Engineer Jun 2023 – Aug 2023

- Designed a normalized data model and modular ETL pipeline to integrate time-series data from 18 disparate IoT sensors into an analysis-ready dataset, processing ≈3 million records across 2 years of hourly readings; reduced pipeline latency by 30% through parallelized processing and optimized API batching.
- Consolidated scattered Python and R scripts into a reproducible R workflow with documentation, enabling non-technical team members to modify and rerun the pipeline independently.

Projects

Next-Step – Full-Stack School Management Platform | GitHub

- Designed a cross-platform REST API with role-based access control for 3 user roles using async Python (FastAPI, SQLAlchemy 2.0).
- Containerized with Docker and deployed via multi-worker uvicorn, validated at 100 concurrent users with 7,801 requests served, zero errors, and 95% of responses under 200ms.
- Integrated Google Drive API and OAuth 2.0 authentication pipeline enabling secure file storage, permission management, and real-time data synchronization across iOS, Android, and Web clients.

ENCoM – Real-Time Generative Speech-to-Speech System | GitHub

- Designed and ran experiments comparing a Mimi+Mamba hybrid architecture against a controlled cascaded baseline, achieving 2.79x inference speedup while maintaining audio quality (MOS) on Switchboard/Expresso evaluation sets.
- Conducted ablation studies on architecture variants, measuring perplexity for same-condition comparisons and automated speech-quality scoring via Whisper transcription for cross-condition evaluation.
- Engineered a low-latency pipeline integrating the Mimi neural audio codec with a Mamba SSM backbone and Temporal/Depth Transformer, reducing time-to-first-audio to <300ms.